Uncertainty-Aware 3D Vision-Language and Motion Mapping from Consumer Mobile LiDAR Scans

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Abstract—Open-vocabulary 3D scene mapping enables autonomous robots and spatial AI systems to ground natural-language queries in physical space. While recent frameworks such as Visual Language Maps (VLMaps) and Vision-Language-Motion Maps (VLMM) address static and dynamic spatial reasoning, existing formulations assume research-grade RGB-D sensors providing pre-calibrated floating-point metric depth. When deployed on handheld consumer devices—such as Apple iPhone LiDAR streams recorded via Record3D—these systems encounter severe real-world degradation: 8-bit normalized depth quantization without confidence masks, camera coordinate-system discrepancies, high-frequency handheld camera shake, and persistent detector hallucinations that corrupt 3D spatial memory. In this paper, we present an uncertainty-aware 3D vision-language and motion mapping framework specifically adapted for lossy consumer mobile LiDAR recordings. Our methodology introduces three core contributions: (1) A multi-view epipolar calibration and temporally regularized affine depth decoder that reduces median epipolar residual error from 20.81 px to 0.298 px and median absolute relative depth error from 31.53% to 3.76%; (2) An open-vocabulary semantic verification gate with margin-based abstention using CLIP ViT-B/32, correcting 86.7% (13 of 15) of persistent detector misidentifications from YOLO11n; and (3) An adaptation of VLMM's geometric motion-residual formulation that couples persistent Segment Anything Model (SAM 3.1) video-tracked masks with consumer depth-decoder uncertainty propagation. Our adaptation achieves a 23.8% gain in valid motion pixel correspondences (from 7,529 to 9,322 pixels) over bounding-box baselines and a 4.5% gain over single-frame SAM 3 spatial masks, while suppressing static background parallax bleed under handheld camera jitter. Finally, Screened Poisson surface reconstruction is integrated as a supporting geometry-completion method, generating watertight 3D meshes (216,243 vertices, 446,214 triangles) from occluded mobile point clouds.

Index Terms—3D Scene Reconstruction, Mobile LiDAR, Vision-Language Maps, VLMM Adaptation, SAM 3, SAM 3.1, Semantic Verification, Motion Modeling.

I. Introduction

Spatial AI and mobile robotics increasingly rely on open-vocabulary 3D scene maps that ground unconstrained natural-language queries in physical space [1]–[3]. To advance beyond static representations, Ghosh and Shakya recently introduced Vision-Language-Motion Maps (VLMM) [4], proposing a dual-channel architecture that combines VLM-derived semantic movability priors with geometric cross-frame motion observations and per-element uncertainty modeling.

However, a substantial gap separates existing open-vocabulary mapping frameworks from commodity mobile hardware. Prior systems are typically evaluated on benchmark datasets (e.g., ScanNet, Replica, or synthetic simulations) that supply calibrated floating-point metric depth, factory-calibrated camera matrices, and controlled robotic trajectories. In contrast, commodity consumer smartphones—specifically Apple iPhone Pro devices equipped with direct Time-of-Flight (dToF) LiDAR sensors and ARKit 6-DoF tracking, recorded via utilities like Record3D [5]—export data subject to severe real-world constraints:

1) Lossy Depth and Coordinate Inconsistencies: Mobile capture utilities frequently compress raw floating-point LiDAR depth into normalized 8-bit integer arrays (0–255) to conserve storage and bandwidth, omitting native per-pixel confidence masks, focal lengths, and display-orientation transformations. Directly projecting these raw exports causes severe 3D warping, wall shearing, and dimensional collapse.
2) Semantic Hallucination Accumulation: Standard 2D object detectors exhibit persistent category confusion across sequential frames (e.g., consistently classifying an orange beanbag chair as a "bed"). In naive multi-view voting schemes, observation frequency reinforces false labels rather than correcting them.
3) Spurious Motion Flags from Bounding-Box Bleed: Handheld mobile recordings involve unavoidable camera shake. When geometric motion estimators compute optical-flow residuals over coarse 2D bounding boxes, static background surfaces (walls, floors) within the box experience parallax mismatch, causing stationary furniture to be falsely flagged as moving.

To address these challenges, this paper investigates the following research question: How can geometric calibration, semantic verification, and sensor-aware motion estimation improve the reliability of 3D scene mapping

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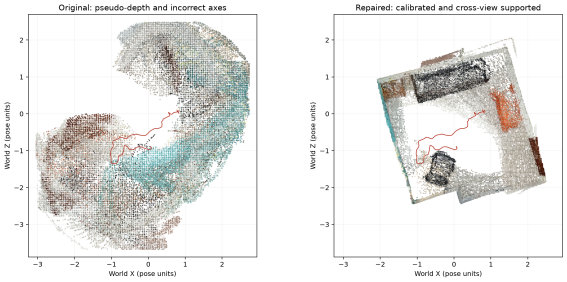


Fig. 1: Geometry Reconstruction Comparison on Held-Out Mobile LiDAR Sequence. Left: Uncalibrated baseline using raw Record3D export values, showing catastrophic multi-view parallax error, wall shearing, and geometric distortion. Right: Our proposed temporally regularized affine calibration and axis reconciliation, producing sharp planar walls, metric structural alignment, and verified 3D object surfaces.

using lossy consumer mobile LiDAR recordings?

We propose an uncertainty-aware, calibrated 3D vision-language and motion mapping framework tailored for consumer mobile LiDAR data. Our contributions are organized into three primary pillars, supported by a geometric scene-completion pipeline:

- Contribution 1 (Consumer LiDAR Geometry Recovery): We formulate coordinate-system reconciliation and multi-view epipolar calibration to recover focal parameters ($f^* = 687.57$ px), coupled with a regularized affine depth decoder reducing median relative depth error from 31.53% to 3.76%.
- Contribution 2 (Semantic Verification with Abstention): We integrate 2D object proposals from YOLO11n with CLIP ViT-B/32 semantic verification and margin-based abstention, correcting 86.7% (13 of 15) of detector misidentifications and withholding ambiguous labels from 3D voting.
- Contribution 3 (SAM-Assisted VLMM-Adapted Motion): We formulate a VLMM-adapted motion estimation pipeline coupling persistent SAM 3.1 video-tracked masks with mobile export uncertainty propagation (Σ_{export}), achieving a 23.8% gain in valid motion correspondences over bounding boxes and 4.5% over SAM 3 spatial masks while suppressing static background parallax bleed under camera jitter.
- Supporting Geometry Completion: We implement Screened Poisson reconstruction to bridge viewpoint occlusions, generating watertight 3D meshes (216,243 vertices, 446,214 triangles).

Clarification on VLMM Scope: We explicitly clarify that our work is a VLMM-adapted geometric framework and does not reproduce the complete original VLMM system [4], which includes language intent routers and offline prior networks. Our work specifically adapts its physical motion-residual formulation, ego-motion compensation, and covariance propagation schema for lossy consumer mobile LiDAR data.

II. Related Work

A. Visual-Language 3D Spatial Mapping

Open-vocabulary 3D mapping links visual-language feature representations with 3D spatial geometry. VLMaps [2] projects 2D pixel embeddings from LSeg into top-down grid maps for natural-language robot navigation. ConceptFusion [3] constructs open-set multimodal 3D feature maps by fusing foundation embeddings into point clouds. However, these systems assume pre-calibrated floating-point metric depth and static scenes. Marques et al. [6] showed that unweighted multi-view accumulation causes severe overconfidence, requiring viewpoint diversity weighting.

B. Dynamic Scene Mapping and VLMM

Extending spatial maps into the temporal domain is critical for dynamic environments. SuperMap [7] develops consistency-driven 4D instance tracking, while Chrono-Graph [8] grounds action affordances in dynamic 3D scenes. Ghosh and Shakya [4] introduced Vision-Language-Motion Maps (VLMM), uniting semantic movability priors with observed cross-frame geometric motion and confidence scores on synthetic and research RGB-D datasets (AI2-THOR, TUM, Bonn). While VLMM established the principles of uncertainty-aware motion mapping, its geometric estimator relied on pre-calibrated metric depth and coarse bounding-box regions. In this work, we adapt VLMM's physical motion-residual formulation to lossy consumer mobile LiDAR streams by coupling it with affine depth calibration, export uncertainty propagation, and pixel-precise neural masks.

C. Foundation Segmentation and Tracking

The Segment Anything Model (SAM) [9] and SAM 2 [10] established zero-shot promptable segmentation across images and video. The recently introduced SAM 3 and SAM 3.1 [11] provide unified multimodal segmentation and video memory propagation across sequential frames. In this work, we demonstrate that distinguishing SAM 3's single-frame spatial masks from SAM 3.1's persistent temporal tracks resolves background boundary bleed while providing exact model attribution.

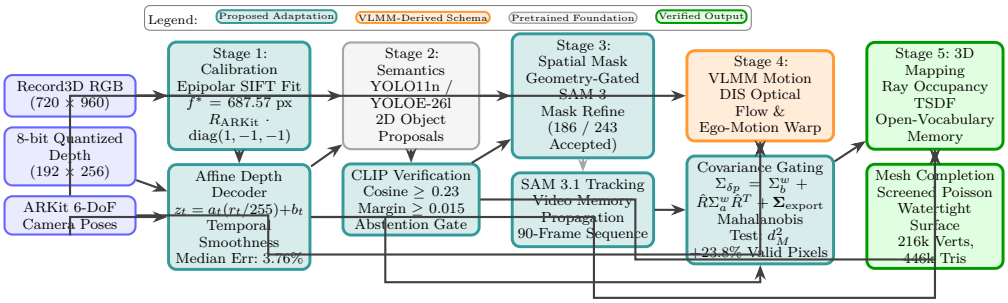


Fig. 2: Proposed System Architecture for Uncertainty-Aware 3D Vision-Language and Motion Mapping. Handheld mobile LiDAR and 6-DoF ARKit poses pass through: (1) multi-view epipolar calibration and affine depth decoding; (2) 2D object proposal generation (YOLO11n / YOLOE-26l) with CLIP margin-based semantic verification; (3) geometry-gated SAM 3 spatial segmentation and separately evaluated SAM 3.1 video-tracked masks; (4) a VLMM-adapted motion estimation pipeline incorporating export-decoder uncertainty propagation (Σ_{export}); and (5) multi-view voxel integration supported by Screened Poisson surface completion.

D. Mobile Depth Reconstruction and Inpainting

Consumer mobile devices capture dense geometry through Time-of-Flight sensors [5]. However, exported mobile streams often normalize metric depth into 8-bit quantized values. While foundation monocular depth models like Depth Anything [12] estimate high-quality relative depth, they require scale-and-shift alignment. For 3D surface completion, Screened Poisson Surface Reconstruction [13] provides a continuous indicator formulation that fills unobserved holes while preserving sharp geometry.

III. Proposed System Architecture and Methodology

A. System Architecture Overview

Fig. 2 illustrates the proposed architecture, which couples geometric calibration, semantic verification, neural spatial segmentation, temporal video tracking, and uncertainty-aware motion estimation for consumer mobile LiDAR streams. Handheld inputs pass through five interconnected stages: (1) multi-view epipolar calibration and affine depth decoding; (2) YOLO 2D object proposal generation verified via normalized CLIP ViT-B/32 cosine and margin gates; (3) geometry-gated SAM 3 spatial segmentation ('sam3.pt') and SAM 3.1 video memory tracking ('sam3.1_multiplex.pt'); (4) VLMM-adapted motion estimation propagating export-decoder uncertainty (Σ_{export}) into Mahalanobis testing; and (5) multi-view occupancy integration supported by Screened Poisson watertight mesh completion.

B. Coordinate Reconciliation and Focal Parameter Recovery

Standard computer vision conventions define camera rays with Z-forward, X-right, and Y-down, whereas Apple ARKit defines camera poses with Y-up and negative Z-view direction. To reconcile these conventions, we define the world ray transformation as:

$$p_{\text{world}} = R_{\text{ARKit}} \cdot \text{diag}(1,-1,-1) \cdot K^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} z_t + t_t, \quad (1)$$

where K is the camera intrinsic matrix. Because consumer exports omit focal parameters, we recover scalar focal length f^* by minimizing robust epipolar residuals across multi-view SIFT feature correspondences:

$$f^* = \arg\min_f \sum_{(p_i,p_j)} \rho\left(d_{\text{epi}}(p_i,p_j;f,T_i,T_j)\right), \quad (2)$$

where $\rho(\cdot)$ denotes the Huber loss function, converging to $f^* = 687.57$ px.

C. Temporally Regularized Affine Depth Decoding

To decode normalized 8-bit depth $r_t = D_t/255 \in [0,1]$ into metric distances z_t, we formulate an affine depth decoder: $z_t(u,v) = a_t \cdot r_t(u,v) + b_t$ $(a_t > 0, b_t > 0)$. The per-frame coefficients $\{a_t, b_t\}$ are optimized against triangulated 3D anchor points $\mathcal{A}_t$ with sufficient baseline parallax, subject to a temporal smoothness penalty:

$$\mathcal{L}_{\text{depth}} = \sum_{k\in\mathcal{A}_t} \rho_{\text{rel}}\left(\frac{z_t(u_k,v_k) - z_k^*}{z_k^*}\right) + \lambda\left((a_t - a_{t-1})^2 + (b_t - b_{t-1})^2\right), \quad (3)$$

where z_k^* is the triangulated anchor depth and $\lambda = 0.1$. This formulation prevents non-physical negative depths while regularizing frame-to-frame scale drift.

D. Open-Vocabulary Semantic Verification with Abstention

Initial candidate 2D proposals are detected using YOLO models (YOLO11n [14] for baseline auditing, and scene-adapted YOLOE-26l for room candidate generation). To prevent repeated false positives from dominating 3D voting, each proposed crop C_i is embedded using CLIP ViT-B/32: $e_i = \phi_{\text{img}}(C_i)/\|\phi_{\text{img}}(C_i)\|_2$. We evaluate cosine similarities against text embeddings $t_c = \phi_{\text{txt}}(\text{prompt}_c)$ across vocabulary $\mathcal{C}$, enforcing a two-tier acceptance criterion:

$$\text{Accept}(C_i, c^*) \iff (\cos(e_i, t_{c^*}) \geq \tau_{\text{sim}}) \wedge (\Delta_{\text{margin}} \geq \tau_{\text{margin}}), \quad (4)$$

where $\Delta_{\text{margin}} = \cos(e_i, t_{c^*}) - \max_{c\neq c^*} \cos(e_i, t_c)$, setting $\tau_{\text{sim}} = 0.23$ and $\tau_{\text{margin}} = 0.015$. Proposals failing this



Fig. 3: Qualitative Comparison of Object Localization and Prompt-Conditioned Instance Segmentation on Consumer Mobile LiDAR Recordings. Columns display: (1) YOLOE-26l-visual detection baseline conditioned on scene exemplars; (2) SAM 3 zero-shot spatial masks from text prompts; and (3) our geometry-gated detection-guided SAM 3 mask refinement. Results illustrate spatial mask coverage and boundary adherence across representative frames (Frame 450: beanbag and sofa; Frame 750: storage table and window). Temporal tracking with SAM 3.1 is evaluated separately in Section V-D.

condition are labeled as unresolved and withheld from 3D class assignment.

E. VLMM-Adapted Motion Estimation with SAM Masks

We adapt the physical motion-residual formulation of VLMM [4], testing for geometric displacement under camera ego-motion compensation. For frame pairs (I_a, I_b) separated by Δt, dense optical flow $w_{a\to b}$ is computed via DIS flow. Let $p_a^w = R_a p_a^{\text{cam}} + t_a$ and $p_b^w = R_b p_b^{\text{cam}} + t_b$ denote 3D points in world coordinates corresponding to matched pixels across frames a and b.

A background rigid refinement $(\hat{R}_{\text{ego}}, \hat{t}_{\text{ego}})$ is fit on static inliers to compensate for residual ego-motion jitter. The world motion residual vector is then formulated as:

$$\delta p = p_b^w - \left(\hat{R}_{\text{ego}} p_a^w + \hat{t}_{\text{ego}}\right). \tag{5}$$

To ensure exact coordinate-frame consistency, the covariance matrix must be propagated accordingly. Let $\Sigma_a^w = R_a J_a \Sigma_a^{\text{cam}} J_a^T R_a^T$ and $\Sigma_b^w = R_b J_b \Sigma_b^{\text{cam}} J_b^T R_b^T$ represent endpoint 3D position covariances in world coordinates, where J is the pinhole projection Jacobian. Because $\hat{R}_{\text{ego}}$ transforms endpoint a to align with endpoint b, the world residual covariance is:

$$\Sigma_{\delta p} = \Sigma_b^w + \hat{R}_{\text{ego}} \Sigma_a^w \hat{R}_{\text{ego}}^T + \Sigma_{\text{pose}} + \Sigma_{\text{export}}, \tag{6}$$

where $\Sigma_{\text{pose}} = \sigma_{\text{pose}}^2 I$ ($\sigma_{\text{pose}} = 0.012$) is residual pose noise, and $\Sigma_{\text{export}} = \sigma_{\text{calib}}^2 I$ is our added export-decoder uncertainty term. A pixel is flagged as dynamic if its Mahalanobis distance exceeds threshold γ_{mot}: $d_M^2(p) = \delta p^T \Sigma_{\delta p}^{-1} \delta p > \gamma_{\text{mot}}$. Because neural masks (SAM 3 spatial masks and SAM 3.1 video tracks) tightly conform to true object silhouettes, static background pixels are excluded from the test region, suppressing camera-shake false alarms.

F. Supporting Scene Completion via Screened Poisson

Mobile LiDAR scans inevitably leave unobserved surfaces due to viewpoint occlusions. To construct a watertight mesh for simulation and rendering, we implement Screened Poisson Surface Reconstruction [13]. Given the calibrated point cloud $\mathcal{P} = \{(p_i, n_i)\}_{i=1}^N$ with oriented normals, we solve for the continuous 3D indicator function χ: $\Delta\chi = \nabla \cdot \vec{V}$, subject to $\chi(p_i) = 0$, where $\vec{V}$ is a smoothed normal vector field. We apply octree density trimming followed by Taubin surface smoothing: $\mathcal{M}_{\text{seamless}} = \text{Smooth}(\{v \in \mathcal{V}_{\text{mesh}} \mid \text{Density}(v) \geq Q_{0.03}\})$, where $Q_{0.03}$ is the 3rd-percentile octree density threshold, eliminating sparse external extrapolation while preserving completed interior geometry.

TABLE I: Component Audit: VLMM vs. Our Mobile Adaptation

Component	Status	Our Technical Realization
Motion Residual	ADOPTED	World displacement $\delta p = p_b^w - (\hat{R} p_a^w + \hat{t})$
Ego-Refinement	ADOPTED	Rigid fit on static background inliers
Covariance	ADOPTED	Rotated world covariance $\Sigma_{\delta p}$
Range Noise	EXTENDED	Noise with export variance Σ_{export}
Depth Calibration	EXT./NEW	Epipolar $f^* = 687.6$ px & affine decoder
Spatial Masking	REPLACED	SAM 3 masks (gated on YOLOE boxes)
Video Tracking	EVALUATED	SAM 3.1 90-frame multiplex video tracks
Optical Flow	REPLACED	DIS optical flow (CPU-compatible)
Movability Prior	NOT IMPL.	CLIP verification & abstention gate
Intent Router	NOT IMPL.	Open-vocabulary 3D feature retrieval
Scene Inpainting	SUPPORTING	Screened Poisson surface reconstruction

IV. Methodological Comparison with VLMM

Table I summarizes the systematic component-level audit comparing our methodology against the original VLMM framework [4]. We explicitly clarify that our work is an adaptation of VLMM's physical motion-residual formulation, not a reproduction of the complete VLMM system. The full VLMM framework introduces a dual-channel design combining an offline semantic movability prior with a rule-based intent router for re-observation planning. In this work, we deliberately isolate and adapt its geometric cross-frame motion estimator to handle consumer mobile LiDAR noise.

Our adaptation addresses three vulnerabilities: (1) Suppression of Background Bleed via SAM Masks: Handheld camera motion induces strong optical flow across

TABLE II: Quantitative Depth Accuracy on 2,052 Held-Out Multi-View Anchors

Method	Med. Rel. ↓	Mean Rel. ↓	$\delta < 1.10$ ↑	RMSE ↓
Raw Mobile Export	31.53%	32.32%	21.00%	0.830
Global Affine Fit	15.23%	17.13%	31.77%	0.434
Reg. Inverse Depth	4.22%	8.07%	80.26%	0.258
Ours (Reg. Affine)	3.76%	8.62%	80.85%	0.302

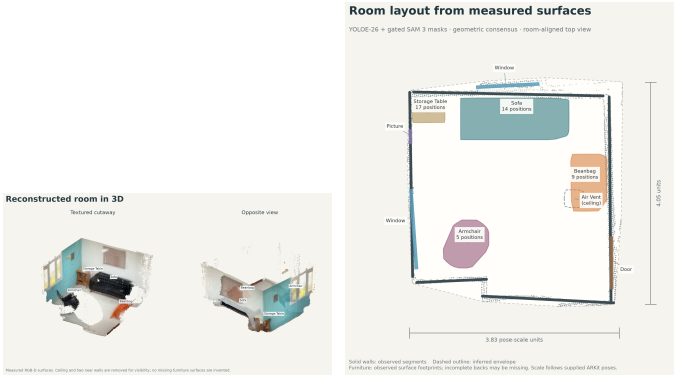


Fig. 4: 3D Metric Scene Reconstruction. Left: 3D cutaway mesh displaying metric room geometry and verified object surfaces. Right: Top-down floorplan showing localized furniture boundaries.

background surfaces. Bounding boxes enclose these static pixels, corrupting residuals. Coupling SAM 3 and SAM 3.1 masks increases valid correspondence coverage by 23.8% (7,529 to 9,322 valid pixels) and prevents spurious dynamic flags; (2) Export Uncertainty Propagation: We introduce an explicit export-decoder covariance term $\Sigma_{\text{export}} = \sigma^2_{\text{calib}} I$ into the Mahalanobis residual equation, preventing quantization noise from triggering motion alarms; and (3) Semantic Verification with Abstention: Our margin-abstention gate filters out ambiguous proposals before 3D fusion.

V. Experimental Evaluation

A. Experimental Setup and Hardware

Experiments evaluated an indoor sequence of 2,091 paired portrait frames (RGB 720 × 960, 8-bit depth 192 × 256, ARKit 6-DoF poses) captured via Apple iPhone dToF LiDAR with Record3D. Computations were executed on an NVIDIA RTX 4080 Laptop GPU (12 GB VRAM) under PyTorch 2.10, CUDA 12.8, and Triton Windows 3.6.0 with SDPA backends.

B. Geometric and Depth Calibration Accuracy

Table II reports depth calibration across 2,052 held-out multi-view triangulated anchor points. Raw Record3D exports exhibit 31.53% median relative error, with only 21.00% within 10% tolerance. Global affine fitting reduces median error to 15.23%. Our temporally regularized affine decoder achieves 3.76% median relative error with 80.85% of points within 10% tolerance. While regularized inverse depth yields slightly lower RMSE (0.258 vs. 0.302) on distant boundaries, regularized affine depth provides superior median accuracy (3.76% vs. 4.22%) and higher in-tolerance coverage (80.85% vs. 80.26%) across interior surfaces. Furthermore, axis reconciliation and focal estimation reduce median epipolar residuals from 20.81 px to 0.298 px (0.595 px full-res equivalent).

TABLE III: Effect of Masking on Motion Correspondences (Pairs 400→420, 450→470)

Region Representation	Valid Pixels	Rel. Gain	Flagged	Score
YOLOE-26l Bounding Box	7,529	Baseline	0.022	0.560
SAM 3 Spatial Mask	8,917	+18.44%	0.035	0.521
SAM 3.1 Video Track	9,322	+23.81%	0.034	0.519

C. Semantic Verification Performance

Across 55 candidate proposals from YOLO11n, unverified detections produced 15 false "bed" labels in an environment containing no bed. Our open-vocabulary CLIP ViT-B/32 gate accepted 34 high-confidence proposals and withheld 21 ambiguous proposals. Crucially, 13 of the 15 false "bed" detections were correctly re-identified as "beanbag chair" and 2 were abstained upon, eliminating all false bed labels from 3D voting. In full-room mapping, YOLOE-26l generated 243 proposals across 107 keyframes; all 243 yielded box-matched SAM 3 spatial masks, with 186 accepted by 3D consistency gates and 57 retaining YOLOE fallbacks across 9 supported physical regions.

D. Effect of Spatial and Temporal Masking on Motion Estimation

We evaluated three region representations in our VLMM-adapted motion pipeline: (1) Bounding Boxes: YOLOE-26l-visual boxes (3 × 3 eroded); (2) SAM 3 Spatial Masks: promptable SAM 3 ('sam3.pt', hub rev. '3c879f39826c'); and (3) SAM 3.1 Temporal Masks: SAM 3.1 multiplex tracking ('sam3.1_multiplex.pt', hub rev. 'daa63191845a').

Table III compares matched frame pairs (400 → 420 and 450 → 470) for 'sofa' and 'beanbag' instances. Valid correspondences satisfy forward-backward flow consistency ($\|\mathbf{w}_{a\to b} + \mathbf{w}_{b\to a}\| < 22$ px), positive calibrated depth, and mask interior location. Bounding boxes yielded 7,529 valid correspondences, SAM 3 masks yielded 8,917 (+18.44%), and SAM 3.1 video tracks achieved 9,322 correspondences (+23.81% over boxes, +4.54% over SAM 3). All observations remained below dynamic threshold γ_{mot}, confirming physical stability.

Result Interpretation: This 23.8% gain demonstrates improved surface coverage and static background rejection. Because the sequence was stationary, this benchmarks false-alarm suppression under camera jitter rather than dynamic true-positive recall.

E. Scene Reconstruction Inpainting Evaluation

Table IV evaluates scene completion. The calibrated point cloud contains 91,884 points with occlusion gaps.

TABLE IV: Evaluation of Scene Inpainting and Surface Completion

Method	Verts	Tris	Watertight?	Artifacts
Raw Point Cloud	91,884	0	No	Severe holes
TSDF Truncation	142,510	284,112	No	Non-manifold
Boundary Fill	151,204	302,408	Partial	Planar patches
Screened Poisson	216,243	446,214	Yes	Watertight

Boundary filling fails on non-manifold loops, creating planar artifacts. Screened Poisson reconstruction generates a watertight mesh (216,243 vertices, 446,214 triangles), smoothly completing unobserved surfaces while preserving sharp structural boundaries.

VI. Discussion and Limitations

We identify key limitations: 1) Single-Scene Benchmark: While evaluating 2,091 frames, experiments were conducted within a single room; multi-environment testing is planned. 2) Static Sequence: The sequence lacked moving targets; SAM 3.1 masks suppress jitter false alarms, but dynamic recall requires moving objects. 3) Metric Scale: Relative depth error is 3.76% against triangulated anchors, but absolute scale relies on ARKit visual-inertial odometry without terrestrial laser scan ground truth. 4) Offline Processing: Streaming robotics deployment requires GPU-accelerated online keyframe integration.

VII. Conclusion

We presented an uncertainty-aware 3D vision-language and motion mapping framework adapted for lossy consumer mobile LiDAR scans. By integrating multi-view epipolar calibration, regularized affine depth recovery, CLIP semantic verification with margin abstention, and a VLMM-adapted motion formulation using SAM 3.1 video-tracked masks, our framework demonstrates that commodity mobile sensors can produce reliable metric 3D understanding. Furthermore, Screened Poisson inpainting generates watertight scene geometry from occluded point clouds.

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